- **Other water management**
  - manage as resources
    - Water agreement
    - Desalination
    - Sustainable water management at local level
      - Water saving
        - change life style — decrease recreational use — replace bath with shower
        - Inplement water saving faucet
      - customer pricing
        - tiered price — reduce water usage
        - private water supply company — offer lower price — increase affordability
      - water purifying — RO — increase water quality
      - zoning
        - setting restriction zone — prohibit sewage disposal — protect environment
        - 新加坡ABC Waters Programme 新加坡“活跃、美丽、洁净”水域计划
          分区目标：将城市水体划分为不同功能区域，平衡生态保护、防洪与公共休闲需求。
          Zoning Goal: Divide urban water bodies into functional zones to balance ecology, flood control, and recreation.
          - 保护区（Protected Catchment）：
            中央集水区（Central Catchment）禁止开发，保护饮用水源与热带雨林。
            Central Catchment Area prohibits development to protect drinking water sources and rainforests.
            法规：禁止排放任何污染物，限制人类活动。
            Regulations: Zero pollutant discharge, restricted human access.
          - 休闲水域（Recreational Reservoirs）：
            滨海湾水库（Marina Reservoir）允许划船、垂钓，但禁止游泳以保障水质。
            Marina Reservoir allows boating/fishing but bans swimming to maintain water quality.
          - 城市排水区（Urban Drainage Zones）：
            通过ABC Waters设计（如碧山公园改造），将暴雨排水渠与公园结合，增强渗透与蓄洪能力。
            Integrate stormwater drains with parks (e.g., Bishan-Ang Mo Kio Park) to improve infiltration and flood resilience.
      - rainwater harvesting
        - 澳大利亚家庭雨水箱 Australian Household Rainwater Tanks
          普及率：43%家庭安装雨水箱（南澳大利亚州达90%）。
          Adoption: 43% households nationwide (90% in South Australia).
          技术：
          屋顶收集→滤网→储水箱（容量2,000-20,000升）→用于冲厕、洗衣与园艺。
          Roof → filter → tank (2,000-20,000L) → toilet/laundry/garden use.
          效益：减少市政用水需求30%，年户均节水10万升。
          Impact: 30% less municipal water use; 100,000L saved per household annually.
      - smart monitoring
  - Other purpose
    - River reclaimation — refill the river channel — to recreate wetland — recover biodiversity — E.g. Kissimmee River
    - IDBM
      - long-term goal of a drainage basin
      - requires involvement of various stakeholders
      - case
        - North Wales — the river Conwy
          - only 27miles long — steep slope — easily flood
          - flow pass Llanrwst — important city
        - resolvement
          - river wall inLlanrwst
          - Build bridge to slow down flow speed upstream — water can spill over for agriculture use
          - downstream low value land — levees
          - raised building and path ways for local residents
        - advantage
          - balancing benefits all stakeholders
          - flood mitigation
          - long term solution
        - disadvantage
          - money consuming
          - time consuming
    - Wetland management